MOHAMMAD TAAN AL AKOUM

AI & Machine Learning Engineer | MSc Artificial Intelligence, CentraleSupélec (Université Paris-Saclay)

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Availability: 6-month end-of-studies internship (stage de fin d'études) from April/May 2027

PROFILE

AI/ML engineer with a mechatronics background (graduated with distinction), completing an MSc in Artificial Intelligence at CentraleSupélec. Research focus: trustworthy multimodal AI for predictive maintenance, from vibration and image data to calibrated, explainable alerts, delivered as reproducible, tested code. Seeking an internship in industrial AI, predictive maintenance, LLM/RAG or autonomous systems.

EDUCATION

MSc in Artificial Intelligence - CentraleSupélec, Université Paris-Saclay, France **2026 - 2027 (expected)**

Coursework: machine learning, deep learning, computer vision, NLP and LLMs, reinforcement learning, MLOps. Spring 2027: advanced deep learning, predictive maintenance, AI planning.

BEng in Mechatronics Engineering - Rafik Hariri University, Mechref, Lebanon **2021 - 2025**

Graduated with distinction, GPA 85.65/100 (French equivalent: mention Très Bien); Dean's and President's Honour Lists.

RESEARCH EXPERIENCE

MSc Research Project - Multimodal Foundation Models for Predictive Maintenance **Oct 2026 - Present**

Laboratoire Génie Industriel (LGI), CentraleSupélec, Gif-sur-Yvette, France - supervised by Zhiguo Zeng

Goal: one model that fuses vibration and force signals with images and text, adapts to new machines from few labels, and issues calibrated, explainable alerts for fault diagnosis, anomaly detection and remaining useful life (RUL).

Code: github.com/taanakoum/Multimodal-AI-for-Predictive-Maintenance | github.com/taanakoum/OmniFuser

- **Reproducibility:** reimplemented OmniFuser (arXiv 2025) as a tested PyTorch package; audited all 35 paper-to-code items, documenting the paper's inconsistent equations and unstated choices; trained it on Ruche A100 GPUs.
- **Evaluation integrity:** found duplicate recordings and label shortcuts in a milling dataset that can outperform the model; designed a leakage-free leave-one-tool-out protocol with schedule and persistence baselines, paired Wilcoxon tests and bootstrap intervals.
- **Architecture:** pipeline on frozen encoders (sensor foundation model, DINOv2, Qwen3-Embedding) with evidential experts for calibrated uncertainty and fusion selection by nested cross-validation, on five public datasets (CWRU, KAIST, XJTU-SY, Nonastreda, MATWI).
- **Diagnosis and explainability:** healthy-only anomaly detection on run-to-failure bearings, scored on alarm onset and false alarms; band-occlusion tests check whether a model relies on the physical bearing defect frequencies.
- **Engineering:** automated tests, GitHub Actions CI (Ubuntu and Windows, Python 3.11 and 3.12), linting and type checking; resumable SLURM jobs tracking every configuration, seed and data checksum.
- **Next:** 15+ papers surveyed; plan for masked and contrastive pretraining, missing-modality tests and LoRA transfer (in progress).

PROFESSIONAL EXPERIENCE

Reliability & Controls Engineering Intern - Ciment de Sibline, Sibline, Lebanon **Aug 2025 - Oct 2025**

- Programmed and validated PLC ladder logic (TIA Portal) for the cooling sequence and gear-rim lubrication of an industrial ball mill, a critical rotating asset whose downtime halts the line.
- Trended SCADA signals (temperature, pressure, flow, drive load) against alarm thresholds, identified early-warning indicators of lubrication, cooling and drive faults, and supported preventive maintenance.

Automotive Diagnostics Intern - Bassoul & Hneine (BMW, Renault), Beirut, Lebanon **May 2025 - Jun 2025**

- Diagnosed vehicle faults from diagnostic trouble codes (DTCs) and ECU data.

PROJECTS

Agentic RAG with Memory - hybrid retrieval, reranking and evaluation suite

- Built an LLM agent (LangGraph, LangChain) that decides when to query long-term memory, call tools or follow up, over a Docling pipeline with structure-aware chunking of text, tables and images.
- Fused TF-IDF and dense retrieval (Sentence Transformers, ChromaDB, MongoDB Atlas) with Reciprocal Rank Fusion and cross-encoder reranking; evaluation suite (Recall@k, MRR, context density) comparing sparse, dense and hybrid retrieval.

Autonomous Smart Walker for Visually Impaired Users - final-year engineering capstone, Rafik Hariri University **2024 - 2025**

- Designed and built the robot end to end (frame, power, sensors, motor control, ROS stack) to guide a visually impaired user indoors.
- Implemented LiDAR SLAM mapping and localisation, path planning with dynamic obstacle avoidance (LiDAR, ultrasonic and camera fusion) and voice commands; ROS on Raspberry Pi, Arduino actuation (Python, C++, OpenCV).

SKILLS

Machine Learning & AI: Python, PyTorch, scikit-learn, NumPy, Pandas, OpenCV; deep learning, computer vision, multimodal fusion, time-series forecasting, anomaly detection, foundation models (DINOv2, PatchTST, TimesFM), LoRA, uncertainty quantification, explainable AI.

LLMs & RAG: LangChain, LangGraph, Docling, Hugging Face, Sentence Transformers, rerankers, ChromaDB, MongoDB Atlas.

MLOps & HPC: SLURM, NVIDIA A100 GPUs, Lightning AI, Linux, Git, GitHub Actions, pytest, Ruff, Pyright, experiment tracking.

Robotics & Industrial: ROS, LiDAR SLAM, sensor fusion, C++, Arduino, Raspberry Pi; PLC (TIA Portal), SCADA, condition monitoring.

Transferable: technical writing and presenting, planning under limited compute, bridging plant-floor and ML teams.

Languages: English - C2 (IELTS); Arabic - native; French - A2, working towards B2 by the start of the internship.

REFERENCES

Available on request.